Simulation Rebuild Protocol v3.2.2

Settled implementation specification — proposed for freeze

Status: This document and its JSON companion define the proposed implementation. No simulation has been run and no scientific result is implied. Freeze only after code-independent review and unit-test design.

1. Settled points from the review

• Noise-only control: true latent signal is X(t)=0 at all times. Each observation is independent measurement error N(0,0.1²). Apply the primary two-consecutive-low-measured-values trigger to these noisy measurements; the observation-rate schedule can therefore respond to noise alone.

• State-space OU surrogate: fit the stationary OU plus independent Gaussian measurement-error model separately to each complete analysed series at its actual timestamps. Use constant mean equal to the series mean, configured θ and ρ grids, and profile the positive diffusion variance Q.

• Define ρ exactly as R/V∞, where R is measurement-error variance and V∞=Q/(2θ) is stationary latent variance. Thus R=ρV∞. For each timestamp gap Δ, use the exact transition Xnext=μ+exp(−θΔ)(X−μ)+η, with η variance V∞[1−exp(−2θΔ)]. Generate the latent OU at the actual timestamps and add independent N(0,R) errors.

• Invalid surrogate statistics count as exceedances; report the invalid count. If all are invalid, set p=1. The Monte Carlo p-value retains the plus-one correction and counts ties as exceedances.

• Random streams: add separate named streams for thinning, each of the 10 subsampling variability draws, and each arm/analysis/null surrogate replicate. Bootstrap uses a cohort-level generator keyed by phase-block seed and model/comparison name, not a patient seed. Record all full stream identities and keys.

• Fold process: continue through T=100; after t=75 hold r=0.05. Check X<−sqrt(r) at every grid time through T and stop a tipped trajectory immediately.

• Fixed-time windows: 43 equally spaced start times, including 0 and T−W. Never-escalated patients use the full series for pre-escalation analysis and are flagged/reported separately.

• Smooth oracle schedule: use each model's own reference level. OU models use their specified μ(t); the fold model uses the upper stable equilibrium x_ref(t)=+sqrt(r(t)).

• Newcombe cross-check: implement the 1998 matched-pair score interval for paired proportions, not the independent-proportions method. Unit-test against published worked examples. The paired patient bootstrap remains primary.

2. Explicit OU surrogate likelihood and generation

For each candidate θ and ρ, fit the exact irregular-time Gaussian state-space model. Let Q be the profiled diffusion variance, V∞=Q/(2θ), and R=ρV∞. The latent process is stationary at constant μ equal to the analysed series mean. Maximise the Kalman log-likelihood over Q>0 at each grid point; choose the (θ,ρ,Q) combination with the largest likelihood. Simulate at the patient's exact timestamps using exact OU transitions, then add independent measurement error. Re-run the identical analysis statistic.

This fitted stationary null is intentionally applied to the complete analysed series. It may absorb some transition structure; that is a limitation to disclose, not a reason to change the method after seeing results. Report AR(1) and shuffle candidate-null results alongside it.

3. Fold process

```
q(t) = clip((t-15)/60, 0, 1)
r(t) = 1 + (0.05-1)*q(t)  for t <= 75
r(t) = 0.05                for t > 75
dX = (r(t) - X(t)^2) dt + 0.1 dW(t)
X(0) = 1.0
At first grid time in [0,100] with X < -sqrt(r(t)): mark tipped and stop.
```

4. Additional random-stream identities

Patient-level streams use patient seed as entropy and the SHA-256-derived key of the full stream name. Thinning: thin|schedule|m|trigger. Variability subsampling: subsample|draw1 through subsample|draw10. Surrogates: nested key (K('surrogates'), K('arm|analysis|null'), k), where k is the zero-based surrogate replicate. Bootstrap: cohort-level entropy equal to the phase-block base seed, with key K('bootstrap|model|comparison'). No global/unrecorded RNG is permitted.

5. Freeze gate and execution order

Before production, test: noise-only control; exact OU surrogate transition and likelihood; rho definition; invalid-surrogate handling; all named random streams; deterministic reproducibility; fold stopping after tipping and post-75 continuation; fixed-time endpoints; never-escalated handling; smooth reference levels; Newcombe worked examples; Phase 1a/1b stop rules; baseline sharing; measurement-error coupling; matched counts. Then run the plumbing test (discard results), Phase 1a, freeze null selection, Phase 1b, and Phase 2 only if calibration passes.

6. Interpretation limits

All simulation findings apply to the specified model family. The smooth schedule is an oracle scenario, not a validated clinician-behaviour model. The clinical MIMIC-IV analysis remains descriptive and cannot confirm the simulated mechanism. Preserve negative results and all candidate-null results.